Jinsook Lee

GitHub | LinkedIn | Website | jl3369@cornell.edu

Google Student Researcher | Socio-technical Systems in Education | Expected Graduation: Aug 2027

Education

2022–present **Cornell University**
Ph.D. in Information Science (Minor in Computer Science)
Committee: René F. Kizilcec (co-chair), Thorsten Joachims (co-chair), Nikhil Garg

2025 *M.S. in Information Science*

Korea University

2019 *M.S. in Fashion Marketing*

2017 *B.S. in Education*

Work Experience

2026 Nov – 2027 Mar **Google** *Student Researcher*
Design and execute a learning impact study on Gemini and NotebookLM
Implement data pipelines and visual logging to extract behavioral insights from student usage data
Supervisor: Dharti Dhami

2025 **NYC FIRST** *Siegel PiTech Fellow*
Analyzed NYC FIRST STEM center data to support low-income students STEM learning experience[blog]
Supervisor: Kate Karageorgiou

2022 **Seoul National University** *Research Fellow*
Built an AI-powered course/major recommender system using historical university enrollment and performance data
Supervisor: Young Hwan Cho, Ph.D.

2019–2022 **Korea University** *Data Scientist*
Developed the first official AI-powered course/major recommender system in South Korea
MOU Project with K-MOOC to develop a course recommender system
Analyzed online learning patterns and course survey results during COVID-19 using LMS dataset
Supervisor: Gyu-Tae Kim, Ph.D.

2019 **Arthur D. Little** *Intern*
Analyzed customer review text data using NLP methods to develop go-to-market strategies for a leading Korean motorcycle manufacturer

Publications

Journal Articles & Conferences

P11 **Jinsook Lee**, Kirk Vanacore, Bakhtawar Ahtisham, Zhuqian Zhou & René F. Kizilcec (2026). What Makes Retrieval Work for Pedagogical Dialogue Annotation? Indexing Granularity over Retriever Adaptation *Conference on Language Modeling (COLM2026)* [Paper]

P10 **Jinsook Lee**, Conrad Borchers, AJ Alvero, Thorsten Joachims & René F. Kizilcec (2026). The Digital Divide in Generative AI: Evidence from Large Language Model Use in College Admissions Essays. *Learning at Scale 2026 (L@S2026)* [Paper] [Inside HigherEd] [NBC News]

P9 Kirk Vanacore, **Jinsook Lee**, Bakhtawar Ahtisham, Sarah Shaw, Justin Reich and Rene Kizilcec (2026). From Tutor Moves to Tutoring States: Modeling the Timing and Sequencing of Pedagogical Strategies for Student Engagement *Learning at Scale 2026 (L@S2026)* [Paper]

P8 Bakhtawar Ahtisham, Kirk Vanacore, Zhuqian Zhou, **Jinsook Lee** and Rene F. Kizilcec (2026). LLM Reasoning Predicts When Models Are Right: Evidence from Coding Classroom Discourse. *Educational Data Mining 2026 (EDM26)* [Paper]

P7 Zhuqian Zhou, Kirk Vanacore, Bakhtawar Ahtisham, Doug Pietrzak, Daryl Hedley, Jorge Dias, **Jinsook Lee**, René F. Kizilcec, Chris Shaw and Ruth Schäfer (2026). Utility-Preserving De-Identification for Math Tutoring: Investigating Numeric Ambiguity in the MathEd-PII Benchmark Dataset. *Educational Data Mining 2026 (EDM26)* [Paper]

P6 Bakhtawar Ahtisham, Kirk Vanacore, **Jinsook Lee**, Zhuqian Zhou, Doug Pietrzak & René F. Kizilcec (2026). AI Annotation Orchestration: Evaluating LLM verifiers to Improve the Quality of LLM Annotations in Learning Analytics. *Learning Analytics and Knowledge (LAK26)* [Paper]

P5 **Jinsook Lee***, Emma Harvey*, Joyce Zhou, Nikhil Garg, Thorsten Joachims & René F. Kizilcec (2024). Ending Affirmative Action Harms Diversity Without Improving Academic Merit. *AAAI/ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO' 24).* [Paper] [Video] * equal contributions

P4 **Jinsook Lee**, Yann Hicke, Renzhe Yu, Chris Brooks, & René F. Kizilcec (2024). The Life Cycle of Large Language Models in Education: A Framework for Understanding Sources of Bias. *British Journal of Educational Technology, 55*(5), 1982–2002. [Paper]

P3 AJ Alvero, **Jinsook Lee**, Alejandra Regla-vargas, René F. Kizilcec, Thorsten Joachims, & anthony lising antonio. (2024). Large Language Models, Social Demography, and Hegemony: Comparing Authorship in Human and Synthetic Text. *Journal of Big Data, 11*(1), 138. [Paper] [Cornell Chronicle]

P2 **Jinsook Lee**, Kibum Moon, Soyeon Han, Sukang Lee, Hyejung Kwon, & Jaeho Han (2021). Development and Application of an AI-Powered Adaptive Course Recommender System in Higher Education: An Example from K University. *Journal of Educational Technology, 37*(2), 267–307. [Paper] [Press]

P1 Kibum Moon, Jinwon Kim, & **Jinsook Lee** (2021). The Early Prediction Model of Student Performance Based on Deep Neural Network Using Massive LMS Log Data. *Journal of the Korea Contents Association, 21*(10), 1–10. [Paper]

Workshop & Non-Archival Conference Publications

W5 **Jinsook Lee**, Conrad Borchers, AJ Alvero, Thorsten Joachims & René F. Kizilcec (2026). The Digital Divide in Generative AI: Evidence from Large Language Model Use in College Admissions Essays. *Stanford Education Data Science Conference*

W4 **Jinsook Lee**, AJ Alvero, Thorsten Joachims & René F. Kizilcec. (2025). Poor Alignment and Steerability of Large Language Models: Evidence from College Admission Essays. *Conference on Language Modeling (COLM25) SoLAR workshop, Social Sim' workshop* [arXiv] [Press]

W3 **Jinsook Lee**, Yann Hicke, Renzhe Yu, Chris Brooks & René F. Kizilcec. (2024). When Bias Meets Personalization: Challenges and Perspectives in LLM-Based Educational Technology. *Learning Analytics and Knowledge (LAK24) Fostering Ethical and Equitable Learning Analytics workshop*

W2 **Jinsook Lee**, AJ Alvero, Thorsten Joachims & René F. Kizilcec. (2024). Application Essays and Characters in Higher Education Admissions. *2024 Annual Meeting of the National Council on Measurement in Education (NCME24)*

W1 **Jinsook Lee**, Bradon Thymes, Joyce Zhou, Thorsten Joachims, & René F. Kizilcec (2023). Augmenting Holistic Review in University Admission using Natural Language Processing for Essays and Recommendation Letters. *Artificial Intelligence in Education (AIED23) EDI in EdTech R&D Workshop* [arXiv]

Book Chapter

B1 Sebastian Munoz-Najar Galvez, **Jinsook Lee**, AJ. Alvero, Sheridan Stewart, René F. Kizilcec, & Amy Desiderio (2025). Artificial Communication and Media Realism in College Admissions. *The Digitized Campus: Artificial Intelligence and Big Data in Higher Education* [Book]

Awards & Grants

2025 **Siegel PiTech PhD Impact Fellowship**
2024 **Cornell Center for Social Sciences Grant**

Invited Talks

2023 Bias in Large Language Models in Education: Sources, Measures, and Mitigation Strategies
National Council on Measurement in Education-AI in Measurement and Education (NCME-AIMC)

2021 How We Upgrade 'AI-advisor' with Graph and the Future of Personalized Recommendation Services in Higher Education
The 7th K.U. Innovation Forum, Seoul, South Korea [video]

Teaching & Service

Teaching Assistant

2022–2026 **Cornell University**
Language and Information (SP26) – *Prof. Cristian Danescu-Niculescu-Mizil*
Learning Analytics (SP25, SP24, FA22) – *Prof. René F. Kizilcec*
Designing Technologies for Social Impact (FA24) – *Prof. Chris Csíkszentmihályi*

Academic Volunteer

2026 **Festival of Learning 2026** *Student Volunteer Chair*
2026 **AIED2026** *LBR Program Committee*

Reviewer

ACM Conference on Fairness, Accountability, and Transparency (FAccT'25 & '26)
Conference on Language Modeling (COLM'25 & '26)
Journal of Learning Analytics (JLA)
International Journal of Artificial Intelligence in Education (IJAIED)

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